

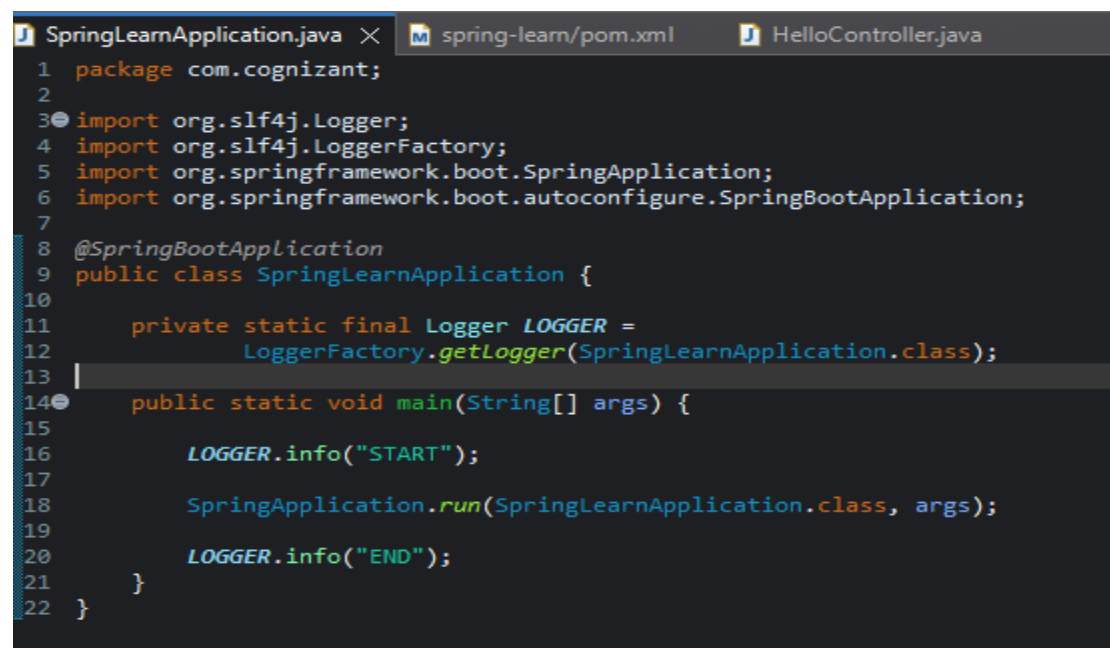
Spring REST using Spring Boot 3

Name: DURGASREE AVVARU

E-Mail: 2300031591cseh@gmail.com

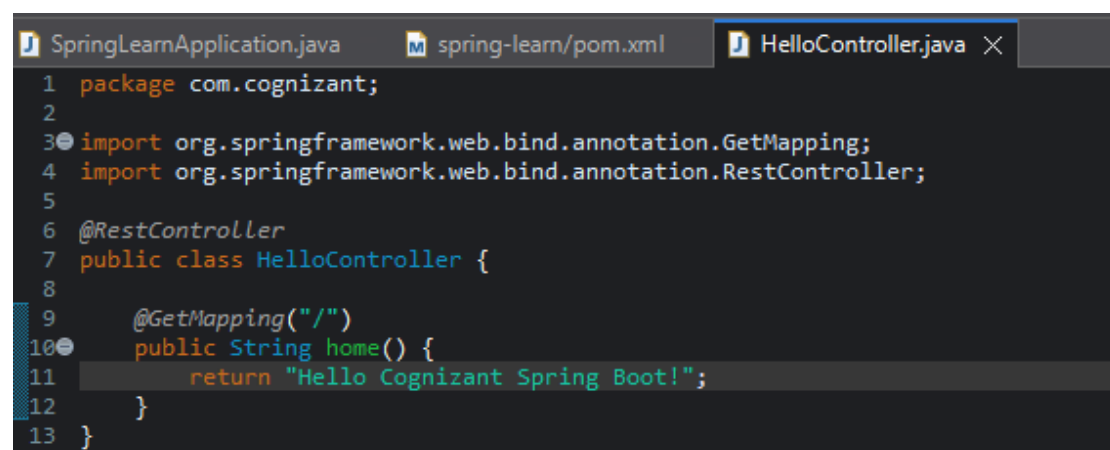
Create a Spring Web Project using Maven:

Spring Learn Application.java:



```
1 package com.cognizant;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.boot.SpringApplication;
6 import org.springframework.boot.autoconfigure.SpringBootApplication;
7
8 @SpringBootApplication
9 public class SpringLearnApplication {
10
11     private static final Logger LOGGER =
12         LoggerFactory.getLogger(SpringLearnApplication.class);
13
14     public static void main(String[] args) {
15
16         LOGGER.info("START");
17
18         SpringApplication.run(SpringLearnApplication.class, args);
19
20         LOGGER.info("END");
21     }
22 }
```

HelloController.java:



```
1 package com.cognizant;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class HelloController {
8
9     @GetMapping("/")
10     public String home() {
11         return "Hello Cognizant Spring Boot!";
12     }
13 }
```

Pom.xml:

```
springLearnApplication.java  spring-learn/pom.xml  HelloController.java
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
4   <modelVersion>4.0.0</modelVersion>
5   <parent>
6     <groupId>org.springframework.boot</groupId>
7     <artifactId>spring-boot-starter-parent</artifactId>
8     <version>4.1.0</version>
9     <relativePath/> <!-- lookup parent from repository -->
10  </parent>
11  <groupId>com.cognizant</groupId>
12  <artifactId>spring-learn</artifactId>
13  <version>0.0.1-SNAPSHOT</version>
14  <name>spring-learn</name>
15  <description/>
16  <url/>
17  <licenses>
18    <license/>
19  </licenses>
20  <developers>
21    <developer/>
22  </developers>
23  <scm>
24    <connection/>
25    <developerConnection/>
26    <tag/>
27    <url/>
28  </scm>
29  <properties>
30    <java.version>17</java.version>
31  </properties>
32  <dependencies>
33    <dependency>
34      <groupId>org.springframework.boot</groupId>
35      <artifactId>spring-boot-starter</artifactId>
36    </dependency>
37    <dependency>
38      <groupId>org.springframework.boot</groupId>
39      <artifactId>spring-boot-starter-web</artifactId>
40    </dependency>
41    <dependency>
42      <groupId>org.springframework.boot</groupId>
43      <artifactId>spring-boot-starter-test</artifactId>
44    </dependency>
45  </dependencies>
46  <build>
47    <plugins>
48      <plugin>
49        <groupId>org.springframework.boot</groupId>
50        <artifactId>spring-boot-maven-plugin</artifactId>
51      </plugin>
52    </plugins>
53  </build>
54 </project>
```

```
SpringLearnApplication.java  spring-learn/pom.xml  HelloController.java
30  <java.version>17</java.version>
31 </properties>
32 <dependencies>
33   <dependency>
34     <groupId>org.springframework.boot</groupId>
35     <artifactId>spring-boot-starter-web</artifactId>
36   </dependency>
37   <dependency>
38     <groupId>org.springframework.boot</groupId>
39     <artifactId>spring-boot-devtools</artifactId>
40     <scope>runtime</scope>
41     <optional>true</optional>
42   </dependency>
43   <dependency>
44     <groupId>org.springframework.boot</groupId>
45     <artifactId>spring-boot-starter-web</artifactId>
46     <scope>test</scope>
47   </dependency>
48 </dependencies>
49
50 <build>
51   <plugins>
52     <plugin>
53       <groupId>org.springframework.boot</groupId>
54       <artifactId>spring-boot-maven-plugin</artifactId>
55     </plugin>
56   </plugins>
57 </build>
58 </project>
```

Output:

```
WEEK-2 TASKS ECLIPSE - spring-learn/src/main/java/com/cognizant/HelloController.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer
LibraryManagement
LibraryManagementExercise2
LibraryManagementExercise4
orm-learn [boot] [devtools]
spring-learn [boot] [devtools]
  src/main/java
    com.cognizant
      HelloController.java
      SpringLearnApplication.java
    src/main/resources
      static
      templates
      application.properties
    src/test/java
    JRE System Library [JavaSE-17]
    Maven Dependencies
    target
    HELP.md
    mvnw
    mvnw.cmd
    pom.xml

SpringLearnApplication.java
spring-learn/pom.xml
HelloController.java

1 package com.cognizant;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class HelloController {
8
9     @GetMapping("/")
10    public String home() {
11        return "Hello Cognizant Spring Boot!";
12    }
13 }

Outline
com.cognizant
  HelloController
    home(): String

Problems Servers Terminal Data Source Explorer Properties Console
spring-learn - SpringLearnApplication [Spring Boot App] (C:\Users\ashu.p2.pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.8.v20230724-1412\jre\bin\java.exe (01-Jul-2026 9:43:06))
21:43:07.002 [main] INFO com.cognizant.SpringLearnApplication -- START
21:43:07.441 [restartMain] INFO com.cognizant.SpringLearnApplication -- START

Spring Boot (v4.1.0)

2026-07-01T21:43:08.399+05:30 INFO 22336 --- [spring-learn] [ restartedMain] com.cognizant.SpringLearnApplication : Starting SpringLearnApplication
2026-07-01T21:43:08.405+05:30 INFO 22336 --- [spring-learn] [ restartedMain] com.cognizant.SpringLearnApplication : No active profile set, falling
2026-07-01T21:43:08.503+05:30 INFO 22336 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults act
2026-07-01T21:43:08.504+05:30 INFO 22336 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related log
2026-07-01T21:43:10.141+05:30 INFO 22336 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8
2026-07-01T21:43:10.166+05:30 INFO 22336 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-01T21:43:10.167+05:30 INFO 22336 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apac
2026-07-01T21:43:10.231+05:30 INFO 22336 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: in

Writable Smart Insert 11:39:291
```

Spring Core – Load SimpleDateFormat from Spring Configuration XML:

Spring Learn HandsOn2 Application.java:

```
*SpringLearnHandson2Application.java X
1 package com.cognizant;
2
3 import java.text.SimpleDateFormat;
4 import java.util.Date;
5
6 import org.slf4j.Logger;
7 import org.slf4j.LoggerFactory;
8
9 import org.springframework.boot.SpringApplication;
10 import org.springframework.boot.autoconfigure.SpringBootApplication;
11
12 import org.springframework.context.ApplicationContext;
13 import org.springframework.context.support.ClassPathXmlApplicationContext;
14
15 @SpringBootApplication
16 public class SpringLearnHandson2Application {
17     private static final Logger LOGGER =
18         LoggerFactory.getLogger(SpringLearnHandson2Application.class);
19
20     public static void main(String[] args) throws Exception {
21         LOGGER.info("START");
22         displayDate();
23         SpringApplication.run(SpringLearnHandson2Application.class, args);
24         LOGGER.info("END");
25     }
26     public static void displayDate() throws Exception {
27
28         ApplicationContext context =
29             new ClassPathXmlApplicationContext("date-format.xml");
30
31         SimpleDateFormat format =
32             context.getBean("dateFormat", SimpleDateFormat.class);
33
34         Date date = format.parse("31/12/2018");
35
36         System.out.println(date);
37     }
38 }
```

Pom.xml:

```
*SpringLearnHandson2Application.java  spring-learn-handson2/pom.xml X
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
4     <modelVersion>4.0.0</modelVersion>
5     <parent>
6         <groupId>org.springframework.boot</groupId>
7         <artifactId>spring-boot-starter-parent</artifactId>
8         <version>4.1.0</version>
9         <relativePath><!-- lookup parent from repository -->
10     </parent>
11     <groupId>com.cognizant</groupId>
12     <artifactId>spring-learn-handson2</artifactId>
13     <version>0.0.1-SNAPSHOT</version>
14     <name>spring-learn-handson2</name>
15     <description>Spring Core Hands-on 2</description>
16     <url/>
17     <licenses>
18         <license/>
19     </licenses>
20     <developers>
21         <developer/>
22     </developers>
23     <scm>
24         <connection/>
25         <developerConnection/>
26         <tag/>
27         <url/>
28     </scm>
29     <properties>
30         <java.version>17</java.version>
31     </properties>
32     <dependencies>
33         <dependency>
34             <groupId>org.springframework.boot</groupId>
35             <artifactId>spring-boot-starter-webmvc</artifactId>
36         </dependency>
37     </dependencies>
38 </project>
```

```
*SpringLearnHandson2Application.java x spring-learn-handson2/pom.xml x
25 <developerConnection/>
26 <tag/>
27 <url/>
28 </scm>
29 <properties>
30 <java.version>17</java.version>
31 </properties>
32 <dependencies>
33 <dependency>
34 <groupId>org.springframework.boot</groupId>
35 <artifactId>spring-boot-starter-webmvc</artifactId>
36 </dependency>
37
38 <dependency>
39 <groupId>org.springframework.boot</groupId>
40 <artifactId>spring-boot-devtools</artifactId>
41 <scope>runtime</scope>
42 <optional>true</optional>
43 </dependency>
44 <dependency>
45 <groupId>org.springframework.boot</groupId>
46 <artifactId>spring-boot-starter-webmvc-test</artifactId>
47 <scope>test</scope>
48 </dependency>
49 </dependencies>
50
51 <build>
52 <plugins>
53 <plugin>
54 <groupId>org.springframework.boot</groupId>
55 <artifactId>spring-boot-maven-plugin</artifactId>
56 </plugin>
57 </plugins>
58 </build>
59
60 </project>
61

Overview Dependencies Dependency Hierarchy Effective POM pom.xml
```

Output:

```
WEEK-2 TASKS ECLIPSE - spring-learn-handson2/pom.xml - Eclipse IDE
File Edit Navigate Search Project Run Design Window Help
Project Explorer X
LibraryManagement
LibraryManagementExercise2
LibraryManagementExercise4
om-learn [boot] [devtools]
spring-learn [boot] [devtools]
spring-learn-handson2 [boot] [devtools]
src/main/java
com.cognizant
SpringLearnHandson2Application.java
src/main/resources
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
HELP.md
mvnw
mvnw.cmd
pom.xml

SpringLearnHandson2Application.java
spring-learn-handson2/pom.xml
Overview Dependencies Dependency Hierarchy Effective POM pom.xml

spring-learn - SpringLearnApplication [Spring Boot App] C:\Users\lokak\p2\poo\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32-x86_64-21.0.8\j21\jre\bin\java.exe (01-Jul-2026 9:43:04)
21:43:07.002 [main] INFO com.cognizant.SpringLearnApplication -- START
21:43:07.443 [restartedMain] INFO com.cognizant.SpringLearnApplication -- START

Spring Boot 1.1 (v4.1.0)
2026-07-01T21:43:08.399+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] com.cognizant.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.8 with
2026-07-01T21:43:08.405+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] com.cognizant.SpringLearnApplication : No active profile set, falling back to 1 default profil
2026-07-01T21:43:08.583+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] com.cognizant.SpringLearnApplication : Devtools property defaults active! Set 'spring.devtools
2026-07-01T21:43:08.594+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] com.cognizant.SpringLearnApplication : For additional web related logging consider setting the
2026-07-01T21:43:10.141+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8880 (http)
2026-07-01T21:43:10.166+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-01T21:43:10.167+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-01T21:43:10.231+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in
2026-07-01T21:43:11.018+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8880 (http) with context path '/'
2026-07-01T21:43:11.021+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] com.cognizant.SpringLearnApplication : Started SpringLearnApplication in 3.57 seconds (process
2026-07-01T21:43:11.030+05:30 INFO 22330 --- [Spring-learn] [ restartedMain] com.cognizant.SpringLearnApplication : END
2026-07-01T21:44:14.869+05:30 INFO 22330 --- [nio-8880-exec-3] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServic
2026-07-01T21:44:14.869+05:30 INFO 22330 --- [nio-8880-exec-3] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServic

Writable Insert 1:1:0
```